



IDBI 2026 INNOVATE

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Team Details

Team Name: CreditSentinel — [Your Team Name Here]

Team Leader: [Your Name Here]

Problem Statement: PS 4 — Default Prediction Model

Brief about the idea

CreditSentinel 360 is a unified, 12-month forward-looking default prediction and early-warning platform for IDBI Bank. It replaces fragmented, structured-data-only scorecards (16-22% accuracy) with a single segment-aware AI engine that fuses structured banking/bureau data with unstructured signals — GST filings, transaction narrations, and news/sector sentiment — to estimate Probability of Default (PD) up to 12 months in advance across Retail, MSME, Corporate and Agri portfolios.

- Every loan, regardless of type, is translated into one common language: a Sentinel Risk Score (0-1000) and Risk Grade (AAA to D).
- Credit teams get a consistent, comparable, and explainable view of portfolio stress — 12 months before it becomes an NPA.
- Built to plug directly into IDBI Bank's sandbox APIs, synthetic datasets, and core banking/bureau feeds.

Opportunities

How different is it from other existing ideas?

- Most existing default models are structured-data-only and fragmented across loan types — separate retail scorecards, separate MSME rules, no common scale.
- CreditSentinel 360 unifies structured + unstructured (NLP-derived) signals into ONE segment-aware model with a common interpretation layer.

How will it solve the problem?

- Forward-looking 12-month PD instead of reactive/lagging DPD-based flags — enables pre-emptive action.
- NLP engine extracts stress signals from GST remarks, transaction narrations and news — catching risk structured data misses.
- Calibrated probability + explainability ensures scores are trustworthy and auditable for credit committees.

USP of the proposed solution

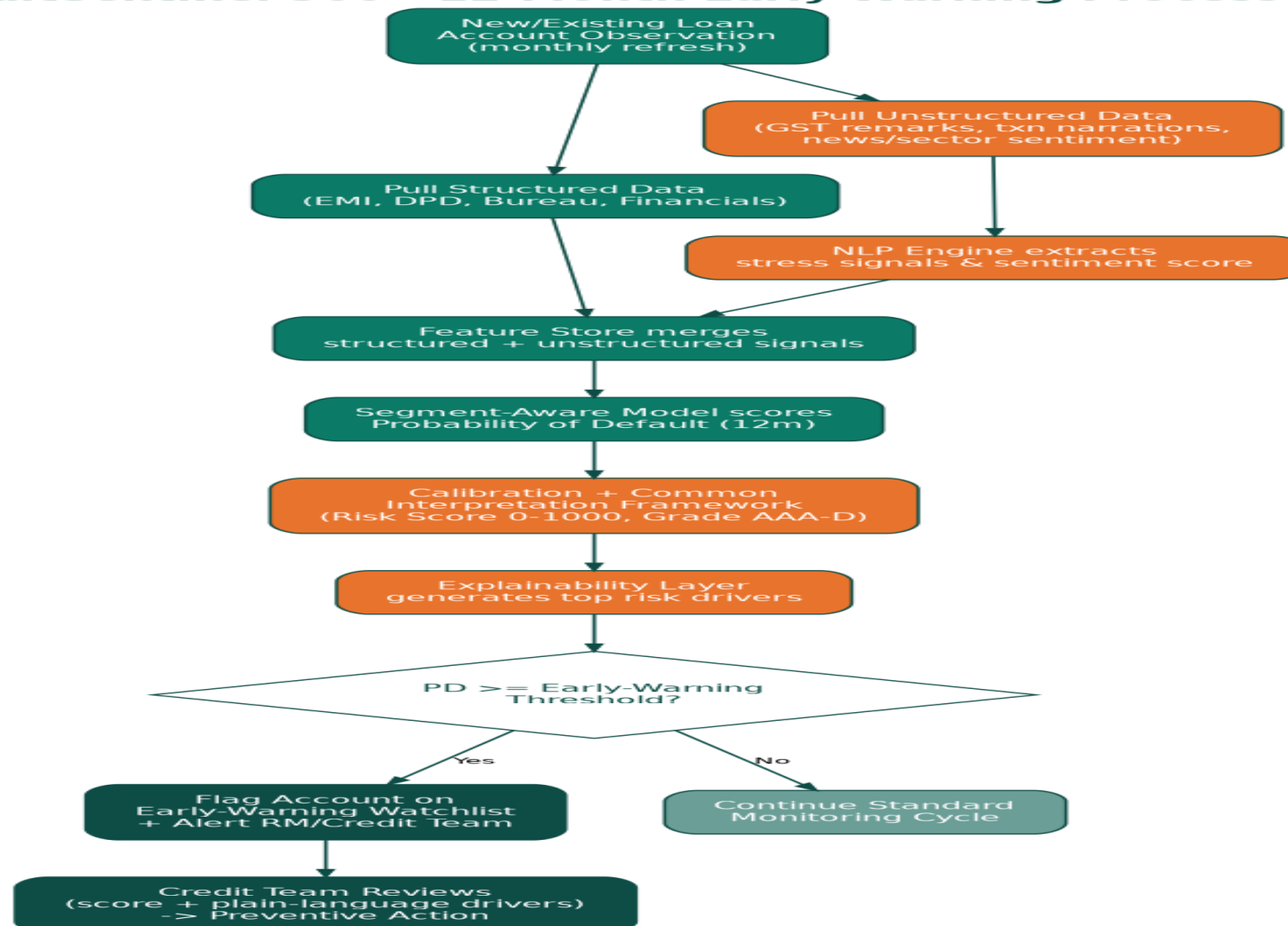
- One score, one grade, one explanation format — usable across Retail, MSME, Corporate and Agri books alike.

List of features offered by the solution

- 12-Month Forward Default Prediction — Probability of Default (PD) estimated a full year ahead, not just current delinquency status.
- Structured + Unstructured Fusion — bureau/EMI/DPD/financial ratios combined with NLP-derived signals from GST filings, transaction narrations and news/sector sentiment.
- Segment-Aware Modeling — a single model conditioned on loan type & borrower segment, avoiding both one-size-fits-all bias and fragmented, hard-to-maintain silo models.
- Common Interpretation Framework — every loan gets a Sentinel Risk Score (0-1000) and Risk Grade (AAA-D), directly comparable across loan types.
- Explainability Layer — top plain-language risk drivers per account, so credit teams see why, not just what.
- Early-Warning Watchlist & Alerts — automatically flags accounts crossing risk thresholds for proactive RM/credit-team action.
- API-First Design — scoring engine callable from Loan Origination and Monitoring systems, ready for sandbox integration.

Process flow diagram / Use-case diagram

CreditSentinel 360 - 12-Month Early Warning Process Flow



Wireframes / Mock diagrams of the proposed solution

CreditSentinel 360

12-Month Forward Default Prediction & Early Warning System | IDBI Innovate 2026 - Problem Statement 4

12,000

Total Accounts

7,411 (61.8%)

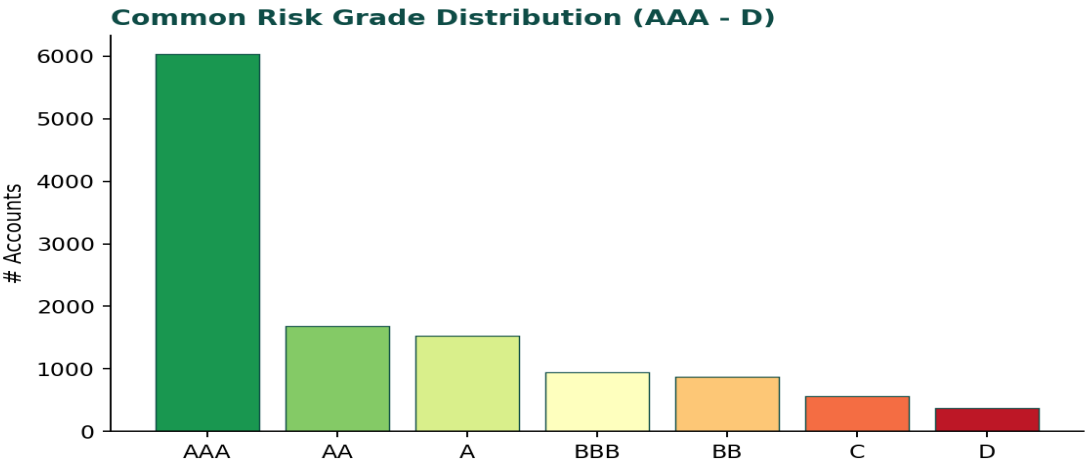
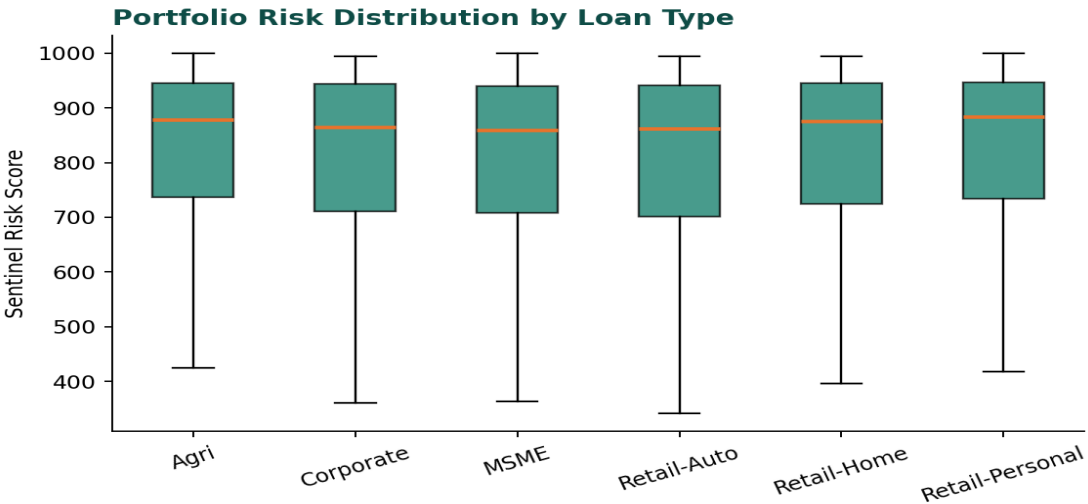
Early-Warning Flags

797/1000

Avg Risk Score

90% (vs 16-22% legacy)

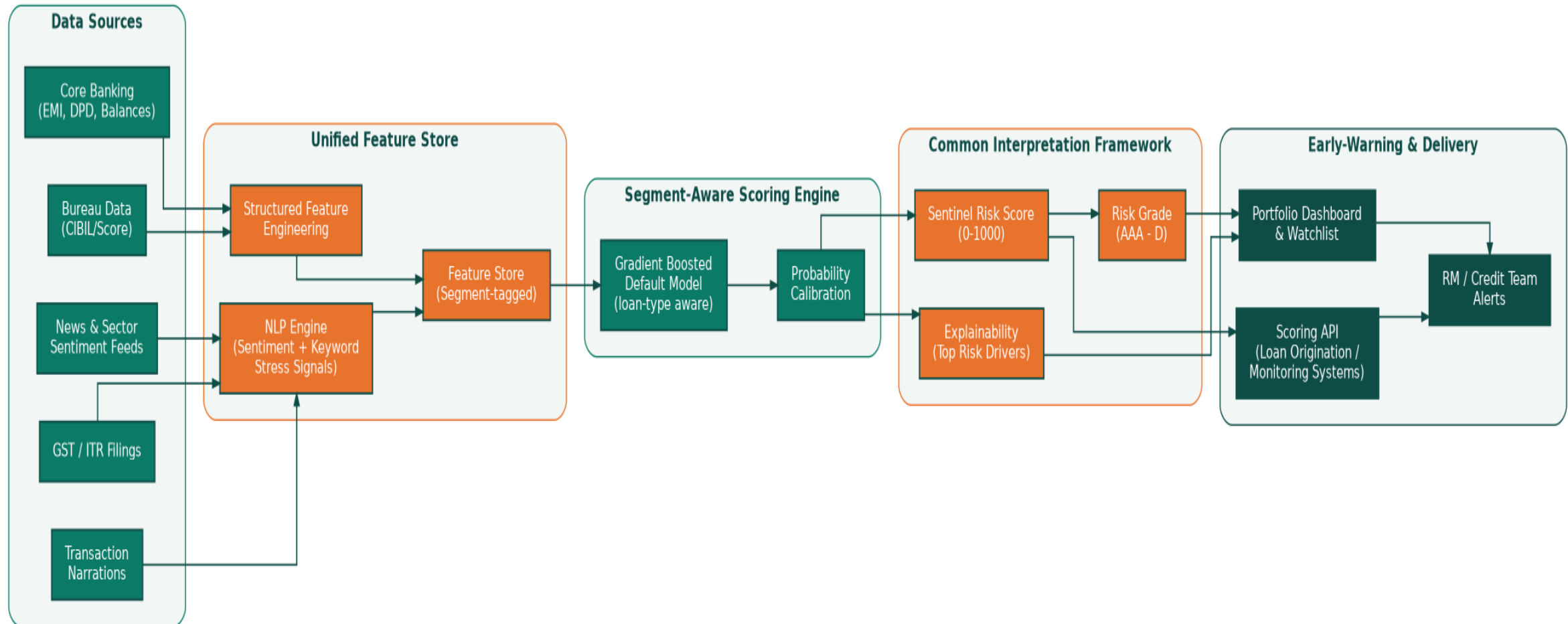
Default-Capture Recall



Loan ID	Loan Type	Risk Score	Grade	12m PD
L108846	Corporate	0	D	100.0%
L108704	Retail-Auto	0	D	100.0%
L101096	Retail-Personal	0	D	100.0%
L104601	Agri	0	D	100.0%
L109404	MSME	0	D	100.0%

Architecture diagram of the proposed solution

CreditSentinel 360 - Solution Architecture



Technologies to be used in the solution

Data & ML

- Python, Pandas, NumPy
- XGBoost (segment-aware gradient boosting)
- scikit-learn (calibration, pipelines)
- Lexicon/NLP sentiment engine for text signals

Explainability & Serving

- SHAP / feature-importance explainability layer
- FastAPI scoring microservice
- Streamlit early-warning dashboard (prototype)

Cloud & Platform (AWS)

- Amazon SageMaker — training & retraining pipeline
- AWS Lambda + API Gateway — scoring API
- Amazon S3 — feature store & data lake
- Amazon QuickSight / CloudWatch — monitoring

Estimated implementation cost (optional)

Phase	Scope	Estimated Cost (INR)
PoC (Sandbox)	Model tuning on IDBI sandbox data, API integration, pilot dashboard	8 - 12 Lakh
Pilot (1-2 business units)	Production-grade deployment, monitoring, RM training	20 - 30 Lakh
Full Rollout	Bank-wide across Retail/MSME/Corporate/Agri, HA infra, MLOps	60 - 90 Lakh
Annual Run-rate	Cloud infra, model retraining, monitoring & compliance	15 - 20 Lakh / year

Indicative, phase-wise estimate — to be refined jointly with IDBI Bank based on sandbox scope, data volumes and infra choices.

Snapshots of the prototype

CreditSentinel 360

12-Month Forward Default Prediction & Early Warning System | IDBI Innovate 2026 - Problem Statement 4

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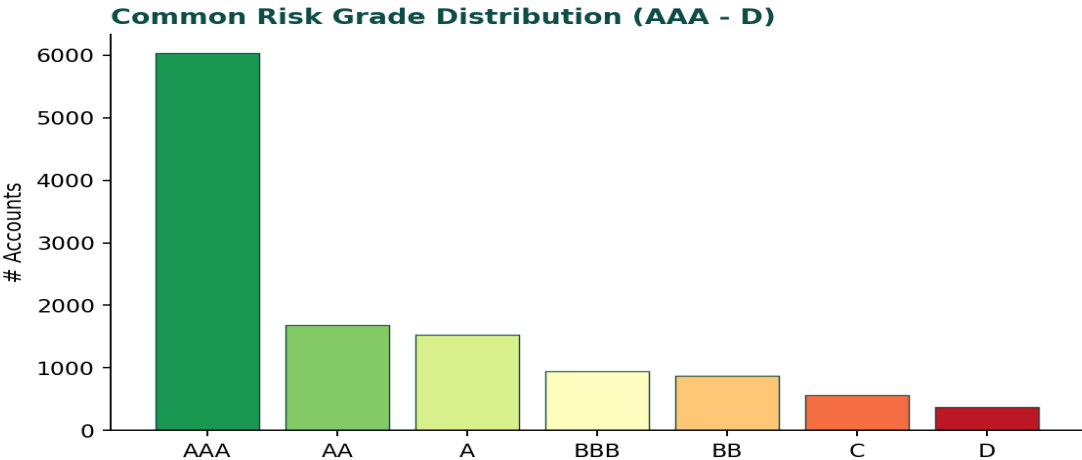
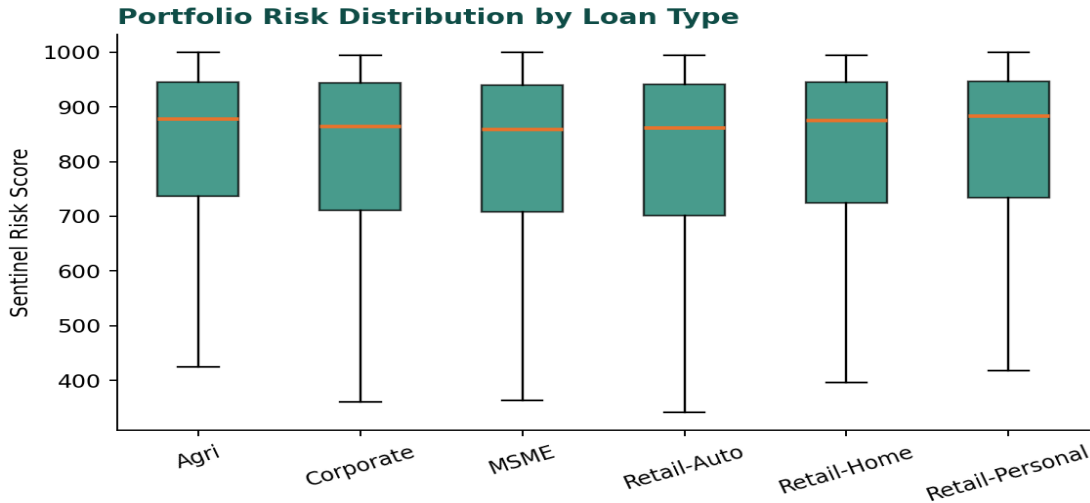
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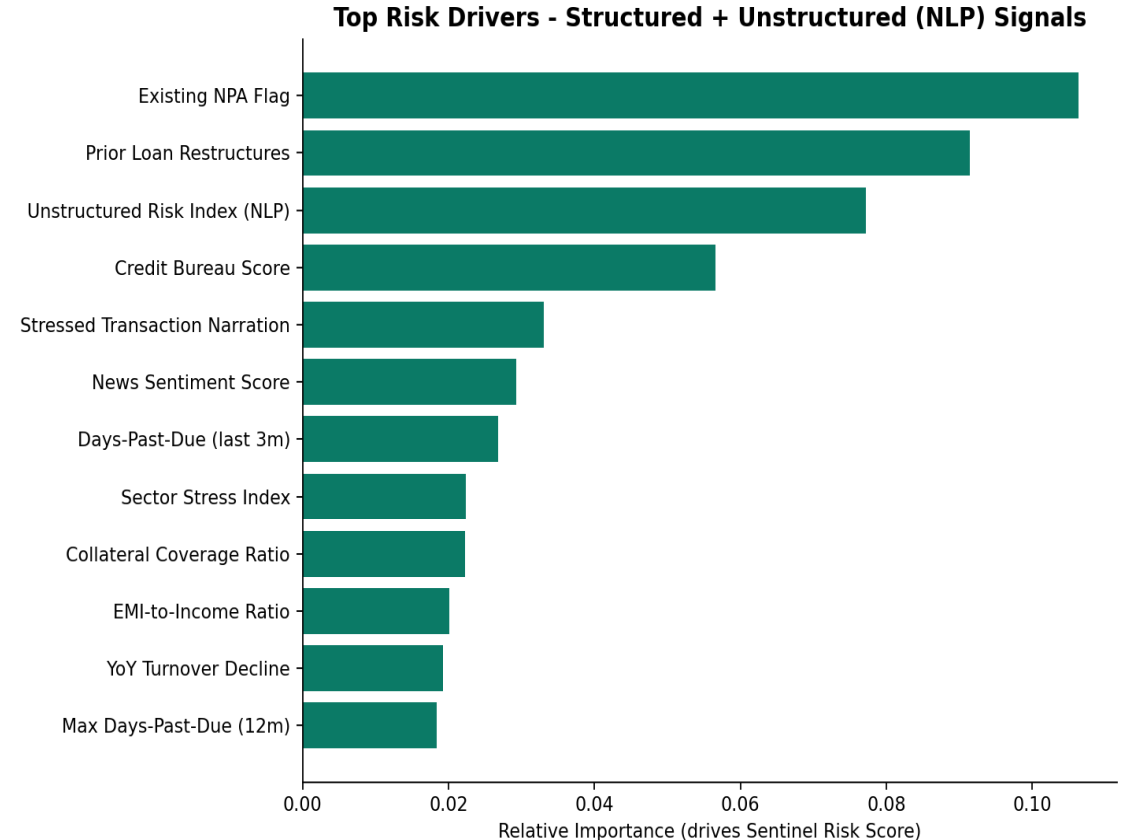


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Prototype Performance report / Benchmarking

Metric	CreditSentinel 360	Legacy baseline
Default-capture recall (12m)	~90%	16 - 22%
ROC-AUC	0.79	n/a (rule-based)
Data used	Structured + Unstructured (NLP)	Structured only
Coverage	Unified across all loan types	Fragmented per loan type

Evaluated on a held-out synthetic test portfolio (n=2,400 accounts). Threshold tuned to maximize precision at ≥90% default-capture recall — directly matching the challenge's target of raising identification accuracy from 16-22% to ~90%.



Additional Details / Future Development (if any)

- Integrate live IDBI sandbox APIs and real bureau/GST feeds to replace synthetic data and retrain on actual portfolio history.
- Add transformer-based NLP (fine-tuned embeddings) for deeper document/news understanding beyond lexicon-based sentiment.
- Introduce macro-economic stress-testing scenarios (rate shocks, sector downturns) for forward PD sensitivity.
- Build a feedback loop where credit officers' outcomes retrain and continuously improve the model (human-in-the-loop MLOps).
- Extend the common interpretation framework into a portfolio-level early-warning report for RBI/regulatory stress reporting.

Provide links to your:

- GitHub Public Repository: [ADD YOUR GITHUB REPO LINK HERE]
- Demo Video Link (3 Minutes): [ADD YOUR DEMO VIDEO LINK HERE]
- Final Product Link: [ADD YOUR DEPLOYED STREAMLIT/APP LINK HERE]



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Thank you!